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AIDEN, a visualization tool in virtual reality for RDF graph

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This paper presents AIDEN (Artificial Intelligence for Data Exploration and Navigation), a system based on virtual reality (VR) to provide an immersive and intuitive application for exploring, analyzing, and modifying RDF knowledge graphs in an interactive three-dimensional environment. The system aims to enhance knowledge capitalization and decision-support processes by helping user to navigate through complex RDF knowledge graphs in an intuitive manner. By facilitating the physical navigation of virtual space with its multi-modal interactions such as gestures, movements, and voice commands; AIDEN allows technical data readily accessible to non-experts. This immersive approach takes advantage of humans' innate ability to interact with three-dimensional environments, thereby enhancing the user experience by reducing cognitive load while exploiting the potential of a third dimension for data visualization purposes. Two distinct knowledge graph representations have been designed to optimize navigation and user experience. The first type, the "overall view," is designed for pattern recognition and group identification. The second, called "immersive view," submerges the user into the graph for a more local perspective, enhancing the readability of individual node relationships. This paper discusses the design and implementation of AIDEN and explores its potential to improve the way to interact with complex data structures.

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Keywords: knowledge management; virtual reality; Design methodology, tools and technologies**1. Introduction**

AIDEN is a system based on virtual reality (VR) to provide an immersive and intuitive application for exploring, analyzing, and modifying knowledge graphs in an interactive three-dimensional environment. It has been designed toward management of historical data with historians.

Aiden offers a voice interface and two navigation modes. One of these takes advantage of the large 3D space in the virtual environment to combine close inspection and global overview.

1.1. Domain-specific issues

The advent of the semantic web has led to the development of complex information structures such as knowledge graphs. In the humanities, there is a widening gap between content experts and the data they work with.

Knowledge graphs based on ontology standards are powerful tools for data sharing, representation, expression and analysis, but they pose significant challenges in terms of consistency checking, navigation and interaction. Traditional 2D interfaces often fail to provide an intuitive user experience. In this case, only a data visualization expert can deal with the content of the graph. Historians are excluded from these practices. In our field of application, consistency checking cannot be limited to automated validations such as shape constraint (SHACL). Information can be true from a logical point of view (when it satisfies the constraints of the ontology), but false from a historical point of view. Then, human-readable content is needed to monitor data quality. This requires the involvement of content experts, the historians in our case. AIDEN aims to facilitate the involvement of historians in the revision of knowledge graphs.

1.1. Proposal

VR training applications have demonstrated an efficient and interactive way of learning and processing new information [1]. While many experiments relies on realistic environment in industrial context [2], we deal with data representations. This paper presents AIDEN (Artificial Intelligence for Data Exploration and Navigation), a system designed to interact with knowledge graphs in a 3D immersive virtual reality (VR) environment (see Fig. 1). This approach takes advantage of humans' innate ability to interact with three-dimensional environments, thereby enhancing the user experience by reducing cognitive load while exploiting the potential of a third dimension for data visualization purposes.

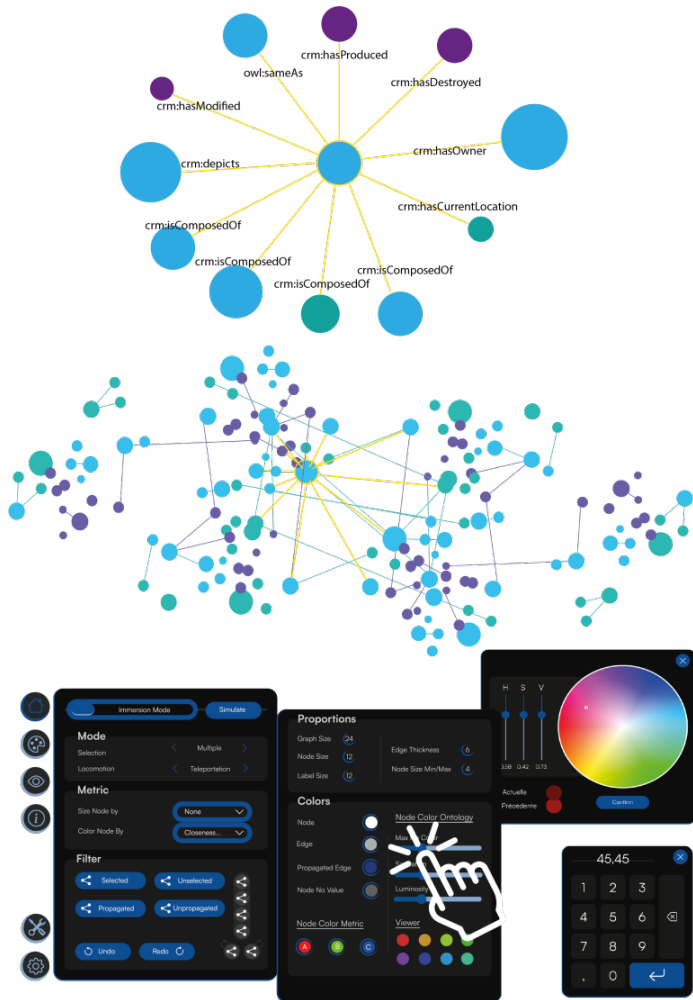


Fig. 1 AIDEN schema of the main UX: In the foreground are the UIs for editing the graph, in the middle is the filtered 3D graph, with the selected node(s) highlighted, and in the background is a detailed 2D graph layout made up of the selected node(s) and their first degree of relationships.

AIDEN incorporates a voice interface to overcome both the difficulties of writing queries and the limitations of software menu displays. In fact, we believe that 2D user experience (UX) overlays should be reassessed for the VR environment. Users can express their requests verbally.

1.2. Scope of application

The graph represents Resource Description Framework (RDF) triples based on standard ontologies. We aim to display graphs from 100 to 800 nodes (i.e. approx. 500-4000 RDF triples, cf. Section 3.1), covering most of the known use cases in our target domain. With less than 100 nodes, 2D applications may be enough to supervise the data; beyond 800 nodes, AIDEN is overloaded and slows down, the display is saturated, unclear. AIDEN only offers to read and update knowledge graphs. Ontology manipulation is out of scope. This means that the underlying semantics provided by the ontology definition (inheritance, transitivity, etc.) are exploited but not displayed. AIDEN is not a tool designed to build and edit ontologies. Only the dataset, i.e. the records, are displayed. The historical records are written by humans and usually come from multiple sources, so they are prone to errors and modeling irregularities.

1.3. Implementation

Development of the 3D environment was carried out with *Unity* game engine. The knowledge graph (historical data) is managed in the *Ontotext GraphDB* tripleStore. *GraphDB* also provides the SPARQL endpoint and the API. Knowledge graphs are built with CIDOC-CRM as core ontology, handling most of the semantics. Additional high level ontologies provide a standard environment (RDFS, OWL, DCterms, etc.). Some other domain specific ontologies such as RDAA refine the description when needed.

The back-end functionality of the voice interface leverages OpenAI APIs (see Fig. 2). The Voice-to-Text conversion is managed by Whisper, while the cognitive processing is executed by ChatGPT.

The application is build and tested with HTC vive focus 3 virtual reality headset.

1.4. Applications

Apart from history and heritage, many disciplinary fields face the same duality of "content experts vs. data visualization experts", with the same technological standards (RDF) and the same amount of data to supervise.

Indeed RDF technologies for data management are widely used in many fields such as biological interactions, industrial performance assessment or global information integration, building information modeling, etc.

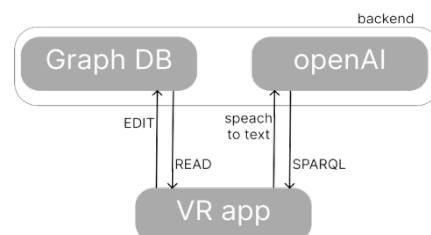


Fig. 2: Global architecture of AIDEN

The emergence of industry 4.0, driven by digital data, connectivity, and cyber-systems, brings the industrial issues closer to the RDF solution and tools.

2. State of the art

2.1. Displaying

Building data visualization charts is a common and powerful way of reading the content of a knowledge graph. Many works such as Li's one [3] are helpful in this sense and offer a complementary approach to AIDEN, which dives into a low-level representation of the data.

The use of 3D graphs in virtual reality (VR) for data visualization has been a topic of interest in recent years. TGView3D [4] highlighted the challenges posed by the size of graphs, which can contain thousands of nodes and an order of magnitude more edges. They suggested the use of three dimensions as a solution, providing more space to organize the graph. In 2008, Ware and Mitchell [5] also emphasized the benefits of 3D graph visualizations, noting that visualizing information structures can lead to easier interpretation. They found that 3D viewing using stereo and motion depth cues significantly increased the size of the graph that could be displayed and read.

2.2. Interacting

Kwon [6] found that spherical layouts and depth routing were generally preferred over 2D graph layouts in an immersive environment. They also found that participants performed better using these techniques than using traditional 2D graph visualization, especially for more difficult tasks and larger graphs. In 2021, Laukka [7] compared hand tracking and controller-based interactions for a VR learning application, and found that hand tracking was seen as more intuitive for users and yielded a higher feeling of presence. However, controllers were seen as more robust. More recently, Rantamaa's team [8] also compared different interaction methods and found that the combination of stylus and controller was the most preferred interaction method.

The use of voice interfaces in VR has also been explored. A study of 2010 [9] found that a multimodal interface was more efficient and users were more satisfied with it when compared to the unimodal interface. Most of the speech input was short phrases rather than complete sentences. In 2019, Hepperle [10] shows that speech interface is favored in ease of learning, uncomplicated handling, speed and input efficiency. The complexity of voice-to-SPARQL translation has been

highlighted [11], the authors proposed a set of specific queries that can be used to interact with the system.

In summary, the literature suggests that 3D graphs in VR offer significant benefits for data visualization. It also suggests that multiple interaction methods, including hand tracking, controller-based interactions, and voice interfaces, can enhance the user experience. Our research aims to contribute to this field in integrating these interaction methods into a single system and providing tools dedicated to common graph operations (filtering, highlighting, tracking issues).

3. Features

3.1. Display

The graph is a directed edge-node multigraph laid out with 3D force-directed algorithm [12]. In order to reduce the number of displayed nodes, only object properties create new edges, while annotation and datatype properties are *attached* to the node and displayed on an overlay panel when the node is selected. Pictures are cast and displayed in the panel.

Users can pin any panel and keep it as a sticky note, positioned wherever they like in the immersive virtual space.

3.2. Voice interface

A voice interface serves a dual role: (1) enabling users to create SPARQL queries and (2) facilitating voice navigation through the application's features. SPARQL query writing is central to RDF graph manipulation, but it requires a good technical knowledge of the query language and is inappropriate for use in a VR environment (virtual keyboard). The voice interface overcomes these issues by providing a simple way to e.g. "hide the nodes of this class [one node is selected]". After some basic local enrichment (selected node data) of the verbal request, chatGPT write a SPARQL request.

3.3. Navigation

Navigation has significant implications for user usability, cognitive load, and spatial awareness. We have developed two distinct knowledge graph representation.

The first one, an overall view of the entire graph, is dedicated to pattern recognition and group identification. The graph is "an object to be handled". Graph rotation and node selection is performed by pre-trained hand gestures. This

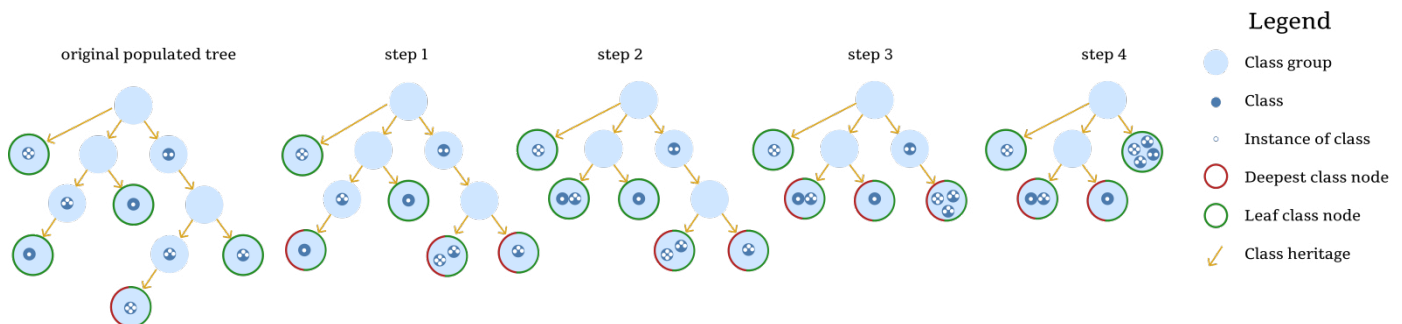


Fig. 3 : Step-by-step algorithm gathering classes within the ontological tree

approach's drawback is a low legibility for the details (text, nodes, and edges). Upon selecting a node, a 2D layout of the node and its direct neighbours is displayed in an enlarged format behind the graph (see Fig. 1), circumventing some legibility concerns. Actually, to inspect the completeness of a very small graph (<20 nodes), a 2D representation is easier to read. If multiple nodes are selected, the background 2D graph may show the path and connection in their first degree neighbourhood. The nodes of this 2D representation are clickable, allowing step-by-step exploration of the graph.

As a contrast, the second representation, immersive view, dives the user into the graph and provides a local view of the data. The graph is "an environment to be explored". This mode enhances the readability of individual node relationships. However, it limits the user's capacity to comprehend the overall structure of the graph. Users navigate via several mechanisms including free-axis flight, node-aimed teleportation, and ground-axis teleportation. As positively evaluated by Hayatpur et al. [13], to assist users in maintaining spatial awareness within the graph, a real-time toggleable mini-map is embedded on the user's virtual wristwatch.

Switching from one mode to another is smoothened by *tweening* the two camera positions; this may help with orientation. Some other navigation shortcuts such as reaching a searched node, backtracking to a previous position augment the overall user experience.

3.4. Graph edition

Our system features a dual-mode. In the Read mode, users can navigate through the graph, identify patterns, and read node information. It is designed to engage the user with the data safely. The voice interface can produce local filters to hide nodes or properties, so to focus the exploration on some part of the data. Toggling to the Edit mode enables users to actively modify the graph database: create, update or delete triples. In this mode, the voice interface may produce INSERT/DELETE SPARQL requests.

3.5. Graph styling

Pattern identification within the knowledge graph is a key tool for the review process. A styling system is implemented to identify areas of interest within the graph. The parameters can be split up into two parts: graph structural metrics styling and ontology class styling.

With structural graph metrics styling, the style of the nodes is influenced by graph theory structural metrics, each providing an insight on the node's position and role within the graph. Classical metrics are implemented: average shortest path length, closeness centrality, betweenness centrality, clustering coefficient and node degree. Each can be mapped to either node color or node diameter.

These metrics are not supposed to be used directly, but can be (vocally) called by the user. For example: "hide nodes with closeness centrality higher than 0.7".

With ontology class styling, the nodes are colored by their class, providing an understanding of their semantic meaning and classification, complementary to their position and connections within the graph.

As many different classes are in use, a mapping is needed to reduce the color palette, making them easier to distinguish. According to Muller [14], for non-regular spatial patterns, cartographers use seven (7) divergent color classes maximum. So we gather node classes based on their positions in the ontologies trees.

This phase runs in the ontology classes' heritage space. As shown on Fig. 3, the algorithm iteratively uplifts the most precise class, moving it up the tree towards its parent class. At each step, among the leaf nodes, the deepest ones are selected, among the deepest ones, the straightest merge is realized. This process loops until the target number of distinct classes is reached; default is seven (7). An optimal color palette is created with ColorBrewer2 [15]. We plan to develop further, so to have distant classes colored with distant colors. We plan to use Wang's algorithm [16].

4. Conclusion and perspectives

1.2. Evaluation

So far, AIDEN testing has been carried out by data-visualization experts aware of 2D graph editing (Cytoscape) in order to enhance this 2D experience.

We plan to evaluate AIDEN with historians (content-experts) in the completion of a number of tasks. The tasks are: (1) create a graph that visualizes the main axes of the dataset, by removing certain nodes with too high centrality; (2) identify inconsistencies or oddities in the graph, e.g. disconnected sub-graphs, missing edges, isolated nodes; (3) stylize the graph to highlight a new pattern; (4) to update the graph: add/remove an edge or a property. The reference time for each task will be established by an expert in graph manipulation, operating without AIDEN (in 2D, on screen). After a short training session, the time taken by content specialists (historians) to complete the tasks using AIDEN will be compared with the reference time.

1.3. Conclusion

As the reliability of knowledge graph content is playing a leading role in the humanities field, its supervision process is a challenge to face. AIDEN provides a way to dive into a low level representation of the content for accurate edits among complex data structures. This low-level representation is enriched by features in a user-centred experience. Empirical user evaluations, additional features and integration of existing works will further enhance the graph review process.

Acknowledgements

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Appendix A. Video of AIDEN

This link provides a video showing AIDEN in action: youtu.be/iCQoeNPtLzg

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